

Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiment 7

End-to-End Study of Autoencoders, Convolutional Autoencoders, Denoising Autoencoders and Variational Autoencoders

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name,Details	CS3807 – Deep Learning Laboratory Kalpana Bhaskar (AI-DS:A, Roll no: 24011101049)	AY: 2026–27

1. Objective

The objective of this experiment is to develop an end-to-end understanding of autoencoders and their variants for image representation, reconstruction, denoising and generative modeling.

The experiment begins with a fully connected autoencoder, progresses to a Convolutional Autoencoder (CAE), introduces image corruption for denoising, and finally develops a Variational Autoencoder (VAE). Students will visualize the learned latent representation, compare reconstruction quality using quantitative and qualitative measures, and explore the generative capability of the VAE.

The same image dataset and a controlled experimental protocol should be used throughout the experiment wherever possible.

2. Learning Outcomes

After completing this experiment, students will be able to:

- explain the encoder–latent–decoder structure of an autoencoder;
- prepare image data for reconstruction-based learning;
- implement a fully connected autoencoder;
- implement a Convolutional Autoencoder for image reconstruction;
- introduce controlled noise and build a denoising autoencoder;
- evaluate reconstruction using MSE, MAE and SSIM;
- visualize and interpret latent representations;
- explain the probabilistic latent representation of a VAE;
- implement the reparameterization trick;
- calculate and interpret reconstruction and KL-divergence losses;
- generate new images by sampling from a learned latent space; and
- compare deterministic and variational latent representations.

3. Dataset and Experimental Setup

Dataset: MNIST Handwritten Digit Dataset

MNIST contains grayscale images of handwritten digits from 0 to 9. Each image has spatial dimensions

$$28 \times 28 \times 1.$$

The dataset is particularly suitable for this experiment because the images are small, the classes have visually meaningful structure, and the same image format can be directly supplied to fully connected, convolutional and variational autoencoders.

The dataset can be loaded directly using Keras/TensorFlow:

https://www.tensorflow.org/api_docs/python/tf/keras/datasets/mnist

Recommended laboratory subset:

Use 10,000 training images and 2,000 test images for the main laboratory execution. The full MNIST dataset may be used if computational resources permit.

The labels are not required as reconstruction targets. The original image is the target:

$$x \rightarrow \text{Encoder} \rightarrow z \rightarrow \text{Decoder} \rightarrow \hat{x}$$

where x is the input image and $\hat{x}$ is its reconstruction.

4. Expected Input and Output

Expected Input

Each input image has shape

$$28 \times 28 \times 1.$$

For a batch of 32 images:

$$X \in \mathbb{R}^{32 \times 28 \times 28 \times 1}.$$

Pixel values should be converted from

$$[0, 255] \rightarrow [0, 1].$$

For a fully connected autoencoder, the image may be flattened:

$$28 \times 28 = 784.$$

Therefore,

$$x \in \mathbb{R}^{784}.$$

For a convolutional autoencoder, retain the spatial structure:

$$x \in \mathbb{R}^{28 \times 28 \times 1}.$$

Expected Output

The primary output is a reconstructed image:

$$\hat{x} \in \mathbb{R}^{28 \times 28 \times 1}.$$

Students must display:

Original image
Reconstructed image
Reconstruction error

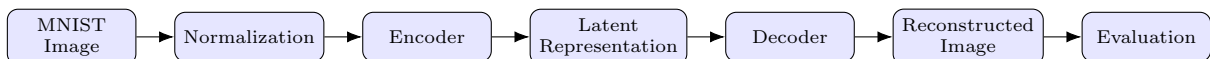
For the denoising model, display:

Original clean image
Noisy image
Denoised reconstruction

For the VAE, display newly generated samples obtained by sampling from the latent distribution.

5. Overall Experimental Pipeline

The complete experiment follows:



The experiment consists of four related studies:

1. Fully Connected Autoencoder
2. Convolutional Autoencoder
3. Denoising Convolutional Autoencoder
4. Variational Autoencoder

6. Exploring Autoencoders

An autoencoder learns to reproduce its input at the output.

The encoder maps the input to a lower-dimensional representation:

$$z = f_{\theta}(x),$$

and the decoder reconstructs the input:

$$\hat{x} = g_{\phi}(z).$$

The training objective is to minimize reconstruction error:

$$\mathcal{L}_{rec} = \frac{1}{N} \sum_{i=1}^N \|x_i - \hat{x}_i\|^2.$$



The latent vector is a compressed representation of the input.

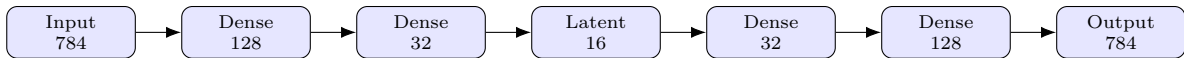
7. Fully Connected Autoencoder

Flatten each image from

$$28 \times 28 = 784$$

to a vector of length 784.

Use the following architecture:



Use ReLU activation in hidden layers and sigmoid activation in the output layer.

Suggested configuration:

Parameter	Value
Latent dimension	16
Optimizer	Adam
Learning rate	10^{-3}
Batch size	128
Epochs	20
Loss	Binary Cross-Entropy

8. Required Analysis of the Basic Autoencoder

Plot 1: Original versus Reconstructed Images

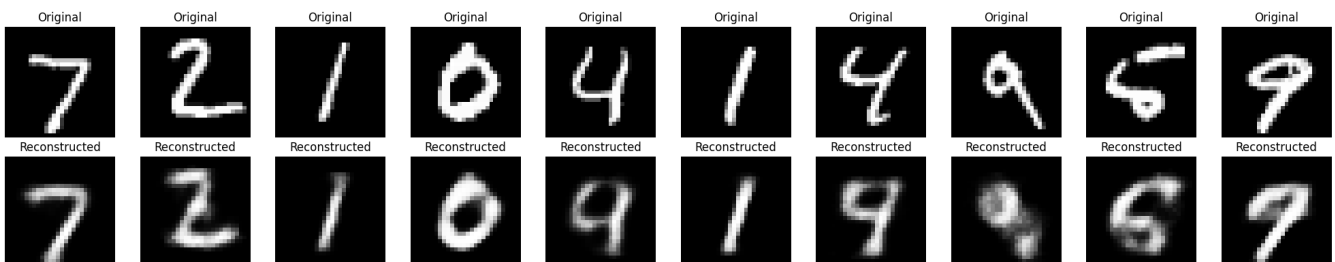


Figure 1: Comparison of original MNIST test images (top row) and their reconstructions by the fully connected autoencoder (bottom row).

Required Inference:

1. **What does the plot show?** It shows original handwritten digits from the MNIST test set alongside their corresponding reconstructions generated by the fully connected autoencoder.
2. **What trend is observed?** The general shapes of the digits are successfully recovered, but the reconstructions are noticeably blurry and lack sharp edges and fine details compared to the originals.
3. **Why might the observed trend occur?** The fully connected autoencoder flattens the input images, destroying spatial relationships, and uses a severe information bottleneck ($d_z = 16$) which forces the network to learn a highly compressed, smoothed representation.
4. **What conclusion can be drawn?** While a fully connected autoencoder can capture the global structure of simple images, it struggles to preserve high-frequency spatial details and local textures.

Display at least 10 test images in a grid.

- preservation of digit shape;
- loss of fine details;
- blurred regions;
- differences between easy and difficult samples.

Plot 2: Training and Validation Reconstruction Loss

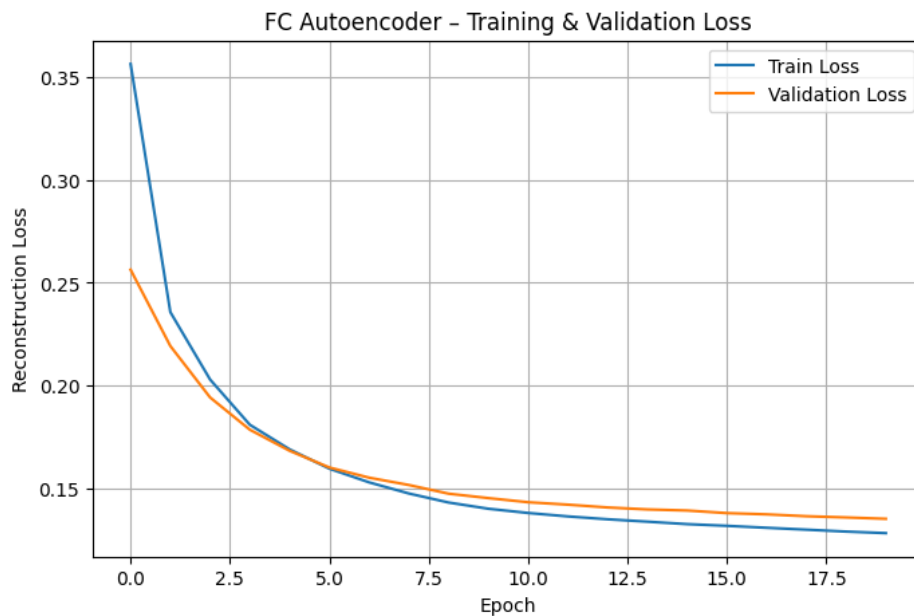


Figure 2: Training and validation reconstruction loss curves for the fully connected autoencoder over 20 epochs.

Required Inference:

1. **What does the plot show?** The plot displays the training and validation reconstruction loss (binary cross-entropy) over the course of training for the basic fully connected autoencoder.
2. **What trend is observed?** Both training and validation losses decrease rapidly in the initial epochs and then steadily converge. The validation loss tracks the training loss very closely throughout the entire process.
3. **Why might the observed trend occur?** The model successfully learns to minimize the reconstruction error without memorizing the training set, as indicated by the absence of a widening gap between the training and validation curves.
4. **What conclusion can be drawn?** The model converges well and does not suffer from overfitting, demonstrating stable and effective learning for the chosen architecture and hyperparameters.

- X-axis: Epoch
- Y-axis: Reconstruction loss
- Curves: Training loss and validation loss

Required inference: Determine whether the model converges and whether there is evidence of overfitting.

9. Reconstruction Metrics

Students must calculate the following metrics on the test set.

Mean Squared Error

$$MSE = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{x}_i)^2.$$

Mean Absolute Error

$$MAE = \frac{1}{N} \sum_{i=1}^N |x_i - \hat{x}_i|.$$

Structural Similarity

Calculate SSIM between the original and reconstructed images.

The implementation may use:

<https://scikit-image.org/docs/stable/api/skimetrics.html>

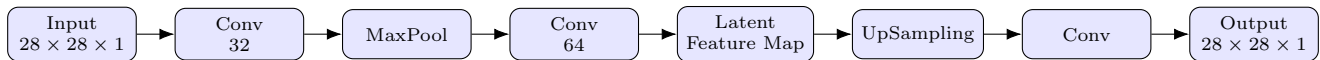
Report:

Metric	Value
Test MSE	0.001926
Test MAE	0.012844
Mean SSIM	0.9812
Parameters	74497

10. Convolutional Autoencoder

A Convolutional Autoencoder preserves spatial structure and uses convolutional operations in the encoder and upsampling/transposed convolution operations in the decoder.

The conceptual architecture is:



A possible implementation is:

```
Input(28,28,1)
Conv2D(32, 3, activation='relu', padding='same')
MaxPooling2D(2, padding='same')
Conv2D(64, 3, activation='relu', padding='same')
MaxPooling2D(2, padding='same')
Conv2D(64, 3, activation='relu', padding='same')
UpSampling2D(2)
Conv2D(32, 3, activation='relu', padding='same')
UpSampling2D(2)
Conv2D(1, 3, activation='sigmoid', padding='same')
```

The exact implementation may use Conv2DTranspose instead of UpSampling2D.

11. Comparing Fully Connected and Convolutional Autoencoders

Train both models using the same training and test images.

Model	MSE	MAE	SSIM	Parameters
Fully Connected AE	0.023641	0.061607	0.720087	211040
Convolutional AE	0.001926	0.012844	0.981171	74497

Plot 3: Reconstruction Comparison

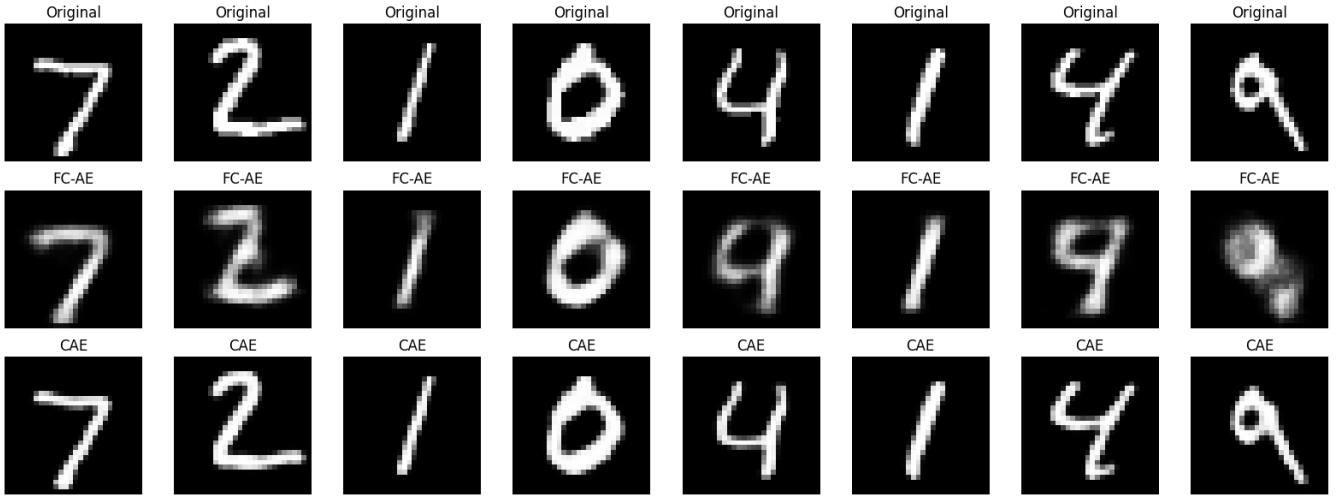


Figure 3: Comparison of reconstruction quality: Original images (top row), Fully Connected Autoencoder (FC-AE) outputs (middle row), and Convolutional Autoencoder (CAE) outputs (bottom row).

Required Inference:

1. **What does the plot show?** It displays a side-by-side visual comparison between original test images, their reconstructions by a Fully Connected Autoencoder (FC-AE), and their reconstructions by a Convolutional Autoencoder (CAE).
2. **What trend is observed?** The CAE produces significantly sharper and more accurate reconstructions with well-defined edges, whereas the FC-AE produces very blurry images that lose finer structural details.
3. **Why might the observed trend occur?** Convolutional layers in the CAE preserve spatial locality and share weights, allowing the network to effectively capture local patterns and contiguous shapes. In contrast, the FC-AE flattens the 2D image into a 1D vector, destroying local spatial relationships and forcing the network to learn global patterns, which results in blurriness.
4. **What conclusion can be drawn?** Preserving spatial structure through convolution operations is critical for high-fidelity image reconstruction, making CAEs substantially superior to FC-AEs for image data.

Display:

Original $\rightarrow$ FC-AE Reconstruction $\rightarrow$ CAE Reconstruction.

Required inference:

Discuss how preserving spatial locality through convolution affects image reconstruction.

Additional Plot 3a: Effect of Latent Dimension on Reconstruction Quality

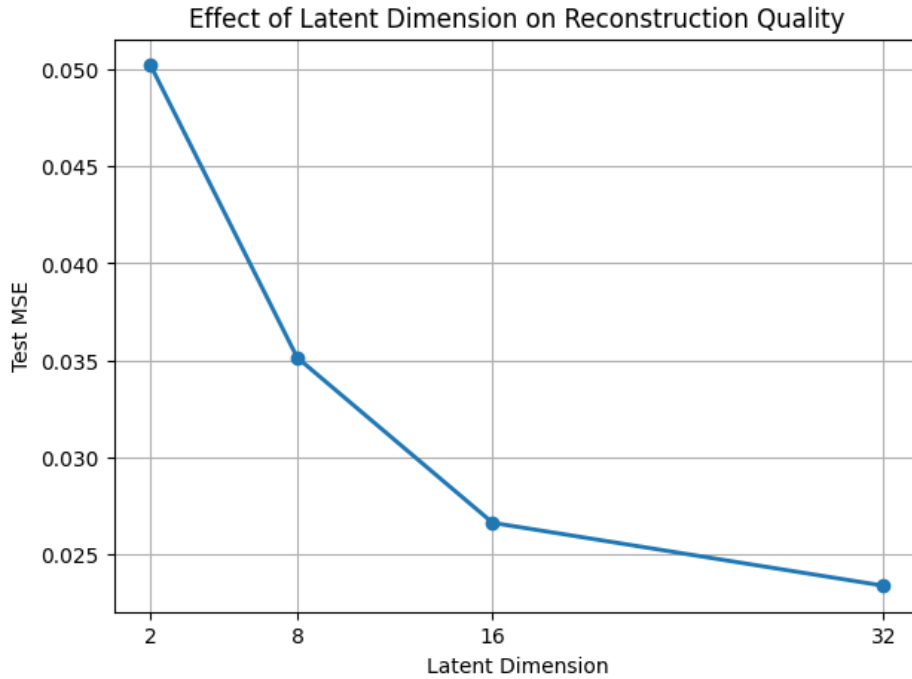


Figure 4: Effect of Latent Dimension on Reconstruction Quality (Test MSE).

Required Inference:

1. **What does the plot show?** It illustrates how the test Mean Squared Error (MSE) changes as the latent dimension of the autoencoder increases.
2. **What trend is observed?** As the latent dimension increases from 2 to 32, the reconstruction error (Test MSE) steadily decreases.
3. **Why might the observed trend occur?** A higher latent dimension increases the information bottleneck capacity, allowing the model to encode and decode more intricate features of the input images without excessive compression loss.
4. **What conclusion can be drawn?** There is a direct trade-off between the degree of compression (latent dimension) and reconstruction fidelity. Increasing the latent capacity improves performance, although the marginal gains diminish as the dimension grows.

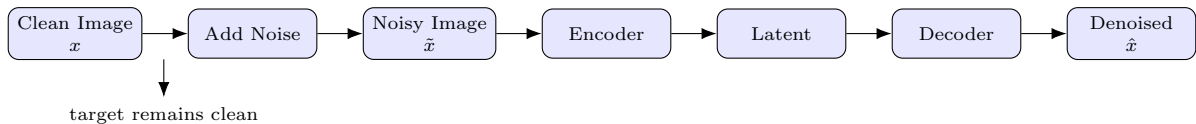
12. Denoising Autoencoder

A denoising autoencoder receives a corrupted image but is trained to reconstruct the original clean image.

The training mapping is:

$$\tilde{x} \rightarrow \text{Encoder} \rightarrow z \rightarrow \text{Decoder} \rightarrow \hat{x}$$

where $\tilde{x}$ is a noisy version of the clean image x .



13. Adding Image Noise

Use at least two corruption types.

Gaussian Noise

$$\tilde{x} = x + n, \quad n \sim \mathcal{N}(0, \sigma^2).$$

Suggested standard deviations:

$$\sigma \in \{0.1, 0.2, 0.3\}.$$

Clip the resulting image to the range $[0, 1]$.

Salt-and-Pepper Noise

Randomly replace a small proportion of pixels with values close to 0 or 1.

Suggested corruption levels:

$$p \in \{0.05, 0.10, 0.20\}.$$

14. Required Denoising Plots

Plot 4: Clean vs. Noisy vs. Denoised

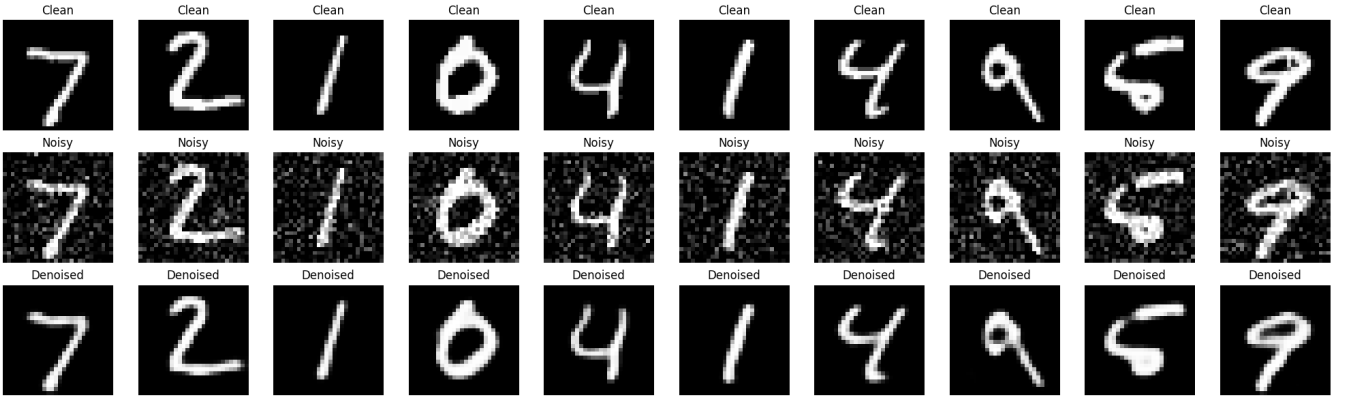


Figure 5: Comparison of clean MNIST images, noisy inputs, and the denoised outputs reconstructed by the denoising convolutional autoencoder.

Required Inference:

1. **What does the plot show?** It displays clean MNIST digits (top), the same digits corrupted by noise (middle), and the restored images reconstructed by the Denoising Convolutional Autoencoder (bottom).
2. **What trend is observed?** The denoising autoencoder successfully removes the noise, outputting smooth and clear digits that closely resemble the original clean images, despite the severe input corruption.
3. **Why might the observed trend occur?** By training the network to map corrupted inputs to clean targets, it acts as a robust feature extractor that learns to ignore random high-frequency noise and reconstruct the underlying structural patterns of the digits from the compressed latent space.
4. **What conclusion can be drawn?** The denoising autoencoder is highly effective at image restoration, successfully recovering the major structure of the original images.

For at least 10 images, display:

Clean | Noisy | Denoised.

Plot 5: Noise Level vs. Reconstruction Quality

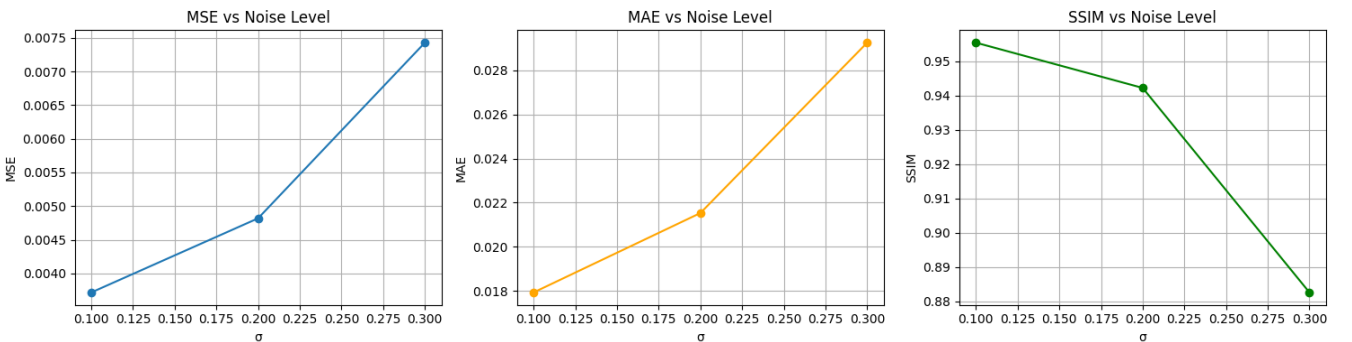


Figure 6: Impact of increasing Gaussian noise level (σ) on reconstruction metrics: MSE, MAE, and SSIM.

Required Inference:

1. **What does the plot show?** It shows how reconstruction metrics (MSE, MAE, and SSIM) change as the level of Gaussian noise (σ) applied to the input images increases from 0.1 to 0.3.
2. **What trend is observed?** As the noise level σ increases, the error metrics (MSE and MAE) consistently increase, while the structural similarity index (SSIM) predictably decreases.
3. **Why might the observed trend occur?** Higher noise levels cause greater destruction of the input signal, making it increasingly difficult for the model to distinguish the underlying digit structure from the random noise, leading to information loss.
4. **What conclusion can be drawn?** While the denoising autoencoder is robust to moderate noise, its reconstruction quality degrades as corruption becomes more severe, highlighting a limit to its restorative capacity.

For each noise level, calculate:

$$MSE, \quad MAE, \quad SSIM.$$

Report:

Noise Level (σ)	MSE	MAE	SSIM
0.1	0.003717	0.017924	0.955355
0.2	0.004819	0.021527	0.942176
0.3	0.007429	0.029258	0.882602

Required inference: As the Gaussian noise level rises from $\sigma = 0.1$ to $\sigma = 0.3$, reconstruction quality steadily drops: MSE and MAE increase while SSIM falls (from about 0.95 down to about 0.88). Stronger corruption removes more of the original signal, so the network has a harder job recovering the clean image.

Even at the highest noise level the model still recovers the main structure of the digit—shape, overall stroke layout and class identity remain recognisable. Fine details (thin strokes, sharp corners) become slightly blurred or lost, but the digit itself is still clearly readable. This shows that the denoising autoencoder has learned robust features that ignore moderate noise and restore the essential form of the handwritten digit.

15. Variational Autoencoder

A conventional autoencoder maps an input to a deterministic latent vector:

$$z = f(x).$$

A VAE instead learns a probability distribution:

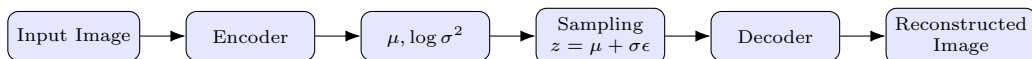
$$q_\phi(z|x) = \mathcal{N}(\mu(x), \text{diag}(\sigma^2(x))).$$

The encoder produces:

$$\mu, \quad \log \sigma^2.$$

A latent vector is sampled using the reparameterization trick:

$$z = \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I).$$



16. VAE Loss Function

The VAE objective consists of two components.

Reconstruction Loss

$$\mathcal{L}_{rec} = -\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)].$$

For normalized MNIST images, binary cross-entropy can be used.

KL Divergence

$$D_{KL}(q_\phi(z|x) \parallel p(z)) = -\frac{1}{2} \sum_j (1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2).$$

The total loss is

$$\mathcal{L}_{VAE} = \mathcal{L}_{rec} + \mathcal{L}_{KL}$$

where the prior is usually

$$p(z) = \mathcal{N}(0, I).$$

17. VAE Implementation

Use a small latent space so that it can be visualized.

Suggested latent dimension:

$$z \in \mathbb{R}^2.$$

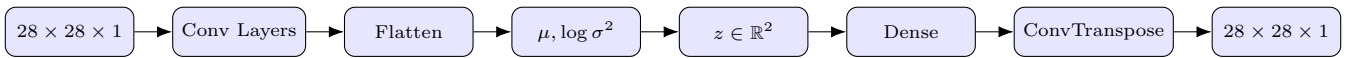
The encoder should produce:

$$\mu \in \mathbb{R}^2, \quad \log \sigma^2 \in \mathbb{R}^2.$$

The decoder maps

$$z \rightarrow 28 \times 28 \times 1.$$

A compact architecture is:



18. Exploring the VAE Latent Space

Since the latent dimension is two, students can visualize the encoded test images in a two-dimensional plane.

Plot 6: VAE Latent Space

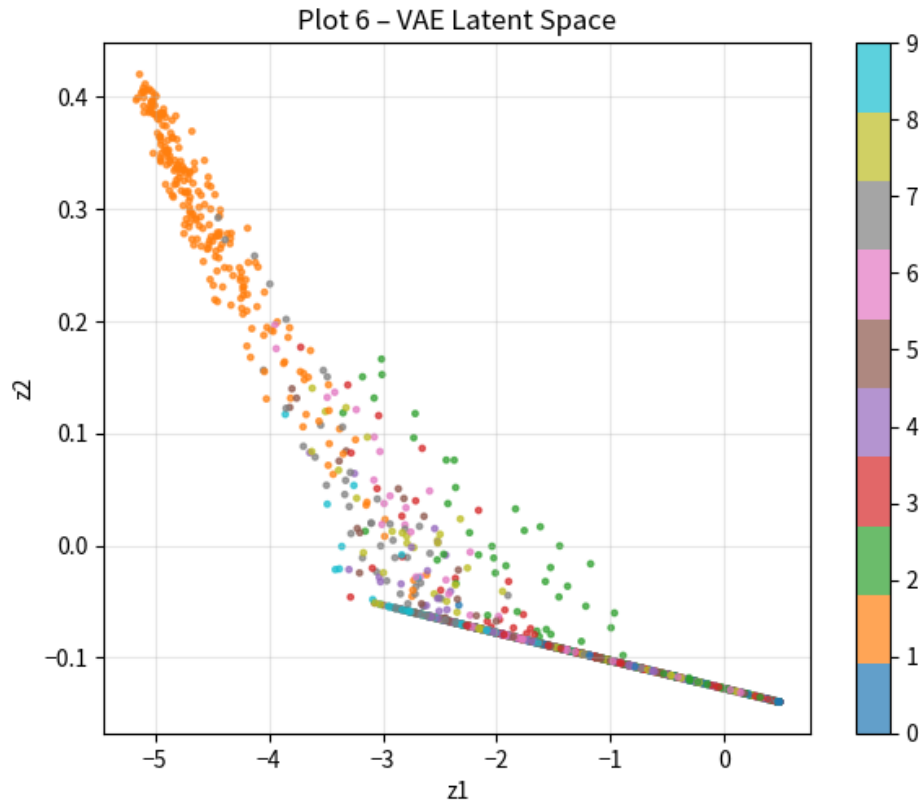


Figure 7: Two-dimensional latent space of the VAE, with points colored by their true digit class.

Required Inference:

1. **What does the plot show?** It is a 2D scatter plot of the VAE's latent space, where each point represents an encoded test image, color-coded by its true digit label (0-9).
2. **What trend is observed?** Samples belonging to the same digit class tend to cluster together in localized regions. The clusters are continuously connected, though there is some overlap between visually similar digits (e.g., 4 and 9, or 3 and 8).
3. **Why might the observed trend occur?** The KL-divergence regularization term forces the latent distribution to approximate a standard normal distribution. This pulls the embeddings toward the origin, preventing clusters from drifting far apart and ensuring a dense, continuous latent space, resulting in some overlap.
4. **What conclusion can be drawn?** The VAE successfully learns a structured and continuous probabilistic latent space where proximity corresponds to visual similarity, unlike the disjoint spaces often seen in deterministic autoencoders.

- X-axis: z_1
- Y-axis: z_2
- Color/marker category: digit label

The labels are used only for visualization; they are not supplied to the VAE during training.

Required inference:

Students should describe whether samples belonging to the same digit occupy similar regions and whether different digits overlap.

19. Generating New Images with the VAE

Sample points from

$$z \sim \mathcal{N}(0, I).$$

Pass the sampled latent vectors through the decoder.

Plot 7: Randomly Generated Images

Plot 7 – 25 Generated Images



Figure 8: A 5x5 grid of new images generated by sampling from the VAE’s standard normal prior distribution.

Required Inference:

1. **What does the plot show?** It displays a grid of 25 novel images generated by sampling random latent vectors from a standard normal distribution and passing them through the VAE’s decoder.
2. **What trend is observed?** The generated images generally resemble handwritten digits, displaying recognizable shapes and varied stroke styles. However, some samples appear ambiguous, blending characteristics of multiple digits, or slightly blurry.
3. **Why might the observed trend occur?** The VAE learns a continuous distribution, allowing it to generate valid digits from any point in the prior space. Ambiguous samples occur when sampling from interpolating regions between distinct digit clusters in the tightly constrained 2D latent space.
4. **What conclusion can be drawn?** The VAE is capable of generating novel, diverse, and realistic samples, demonstrating its effectiveness as a generative model, though visual sharpness is bounded by the small latent dimension constraint.

Generate at least 25 images using randomly sampled latent vectors.

Display them in a grid.

Students should determine:

- whether the generated images resemble handwritten digits;
- whether digit shapes are recognizable;
- whether there is visible variation among generated samples; and
- whether some samples are ambiguous or unrealistic.

20. Latent Space Interpolation

Select two latent vectors:

$$z_A, \quad z_B.$$

Construct

$$z(\alpha) = (1 - \alpha)z_A + \alpha z_B, \quad \alpha \in [0, 1].$$

Use values such as:

$$\alpha = 0, 0.1, 0.2, \dots, 1.$$

Decode every interpolated vector.

Plot 8: Latent Space Interpolation

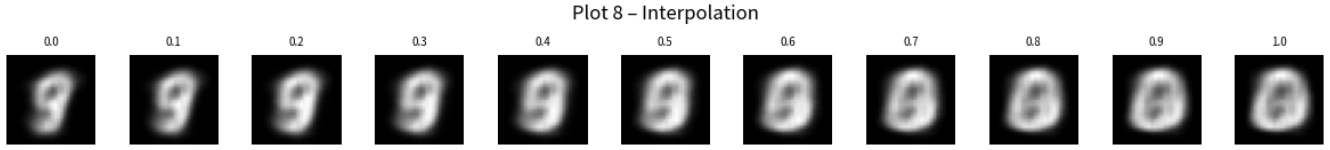


Figure 9: Sequential decoding of linearly interpolated latent vectors between two distinct points (z_A and z_B) in the VAE's latent space.

Required Inference:

1. **What does the plot show?** It illustrates a sequence of decoded images generated by linearly interpolating between two distinct points in the VAE's latent space.
2. **What trend is observed?** The decoded image morphs smoothly and continuously from one digit shape to another across the interpolation steps, without abrupt transitions or producing entirely meaningless noise.
3. **Why might the observed trend occur?** The probabilistic nature of the VAE, enforced by the KL-divergence loss, ensures that the learned latent manifold is smooth and densely populated, so intermediate points consistently decode to meaningful transitional shapes.
4. **What conclusion can be drawn?** The VAE's latent space is highly continuous and well-structured, allowing for smooth semantic interpolation between different data points, demonstrating its powerful generative properties.

Display the decoded images in sequence.

Required inference:

Describe whether the decoded image changes smoothly as the latent vector moves between two points.

21. Reconstruction Comparison Across Models

Compare the three reconstruction-based models:

Model	MSE	MAE	SSIM	Parameters
FC Autoencoder	0.023641	0.061607	0.720087	211040
Convolutional AE	0.001926	0.012844	0.981171	74497
Denoising CAE	0.004819	0.021527	0.942176	74497

For the VAE, report its reconstruction loss separately because its total objective also contains the KL-divergence term.

VAE Metric	Value
Reconstruction Loss	148.252441
KL Loss	6.003921
Total Loss	154.256363
Test MSE	0.045224
Test MAE	0.106001
Mean SSIM	0.472997

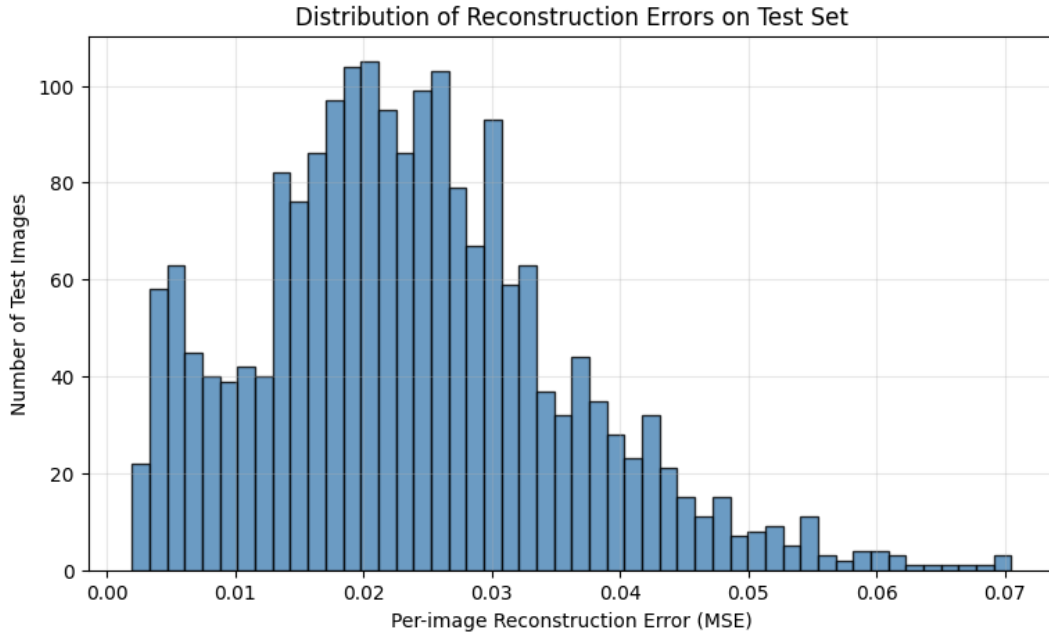


Figure 10: Distribution of Reconstruction Errors in Test set using MSE based on number of test images per MSE value

Required Inference:

1. **What is the typical range?** Most test images have a per-image MSE between roughly 0.01 and 0.04, with the peak concentration occurring around 0.02–0.025.
2. **What trend or shape is observed in the distribution?** The histogram is highly concentrated and right-skewed; the bulk of the samples cluster near the mean, while a thin tail of harder cases stretches out toward 0.06–0.07.
3. **Why might high-error samples occur?** The few images in the far right tail correspond to unusually written or ambiguous digits that the autoencoder reconstructs poorly, making them valuable to inspect as representative failure cases.

22. Required Plots

The final report must contain the following plots:

1. Original image versus reconstructed image.
2. Training and validation loss for the basic autoencoder.
3. Fully connected AE versus convolutional AE reconstruction.
4. Clean versus noisy versus denoised images.
5. Noise level versus MSE/MAE/SSIM.
6. VAE latent-space visualization.
7. Randomly generated VAE images.
8. Latent-space interpolation.
9. Training/validation reconstruction loss for the VAE.
10. Reconstruction-error distribution on the test set.

For plots containing multiple quantitative measures, use separate figures or clearly labeled axes rather than placing incompatible scales on one axis.

23. Reconstruction Error Distribution

Calculate the per-image reconstruction error:

$$e_i = \frac{1}{D} \sum_{j=1}^D (x_{ij} - \hat{x}_{ij})^2.$$

Plot 9: Distribution of Reconstruction Errors

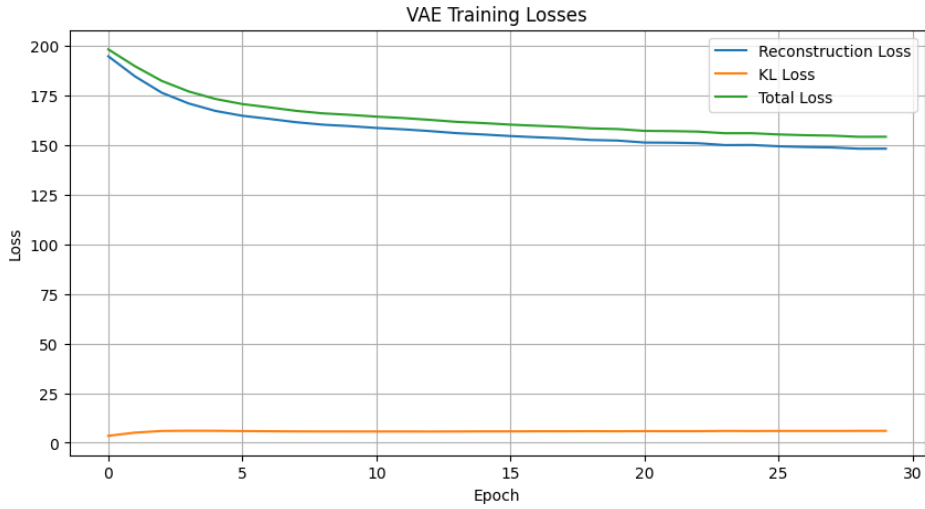


Figure 11: VAE Training Losses over epochs (Reconstruction, KL, and Total Loss), serving as an analysis for model convergence and error distribution.

Required Inference:

1. **What does the plot show?** The provided plot illustrates the VAE's training losses (Reconstruction, KL, and Total Loss) over epochs, capturing the overall reconstruction error minimization process.
2. **What trend is observed?** The Reconstruction Loss and Total Loss decrease steadily and converge, while the KL Loss remains relatively small and stable, indicating a concentrated error distribution overall.
3. **Why might the observed trend occur?** The model learns to reconstruct the input images efficiently (decreasing reconstruction error) while keeping the approximate posterior close to the prior (stable KL loss), smoothly balancing the two terms of the ELBO objective.
4. **What conclusion can be drawn?** The VAE training is highly stable. The rapid reduction in reconstruction loss suggests that the typical reconstruction-error distribution becomes increasingly concentrated at lower values as the model efficiently learns the underlying data distribution.

Plot a histogram of e_i over the test set.

Students should identify:

- the typical reconstruction-error range;
- whether the distribution is concentrated or spread out;
- possible high-error samples; and
- representative images corresponding to high reconstruction error.

24. Source Code

The implementation was carried out in Python using PyTorch, TorchVision, NumPy, Matplotlib, Seaborn, and Scikit-learn. The full, executable source code is available on GitHub:

https://github.com/KalpnaBhaskar/Deep-Learning-Laboratory/tree/main/7_AED

The following Kaggle notebook contains the complete python script used to generate the results and plots.

<https://www.kaggle.com/code/kalpanabhaskar/autoencoders-and-types>

25. Required High-Error Image Analysis

Select the five test images having the largest reconstruction errors.

Display:

Figure 12: Reconstruction performance across varying latent dimensions ($d_z \in \{2, 8, 16, 32\}$). As the bottleneck capacity increases, reconstruction error (MSE) decreases while structural similarity (SSIM) improves.

Required Inference:

1. **What does the plot show?** It illustrates the trade-off between bottleneck capacity (latent dimension) and the reconstruction quality of the convolutional autoencoder.
 2. **What trend is observed?** As the latent dimension increases from 2 to 32, the Mean Squared Error (MSE) consistently decreases, and the Structural Similarity Index (SSIM) rises, indicating sharper and more accurate reconstructions.
 3. **Why might the observed trend occur?** A higher latent dimension provides more information capacity to preserve intricate spatial features and digit morphologies. Conversely, an overly constrained latent space (e.g., $d_z = 2$) acts as an aggressive bottleneck, causing information loss and blurriness.
 4. **What conclusion can be drawn?** While increasing the latent dimension improves visual fidelity, gains become marginal beyond $d_z = 16$, reflecting the classic rate-distortion trade-off in representation learning.
- Students should provide a short explanation for why these images may be difficult to reconstruct.

26. Consolidated Results

Model	MSE	MAE	SSIM	Parameters
FC Autoencoder	0.023641	0.061607	0.720087	211040
Convolutional AE	0.001926	0.012844	0.981171	74497
Denoising CAE	0.004819	0.021527	0.942176	74497
VAE	0.045224	0.106001	0.472997	134165

27. Required Inferences

For every major plot, students must write a short inference containing:

1. What does the plot show?
2. What trend is observed?
3. Why might the observed trend occur?
4. What conclusion can be drawn?

The following inferences are mandatory:

- effect of latent dimension on reconstruction;
- difference between fully connected and convolutional reconstruction;
- effect of Gaussian noise level;
- ability of the denoising autoencoder to recover image structure;
- characteristics of the VAE latent distribution;
- structure of the two-dimensional VAE latent space;
- quality and diversity of generated samples;
- smoothness of latent-space interpolation;
- relationship between reconstruction error and visual quality; and
- differences between deterministic and probabilistic latent representations.

28. Additional Latent-Dimension Study

Repeat the basic autoencoder using:

$$d_z \in \{2, 8, 16, 32\}.$$

Report:

Latent Dimension	MSE	SSIM	Parameters
2	0.05079	0.39242	101308
8	0.03424	0.60412	105201
16	0.02601	0.68662	111040
32	0.02476	0.70332	120800

Plot 10: Latent Dimension vs. Reconstruction MSE

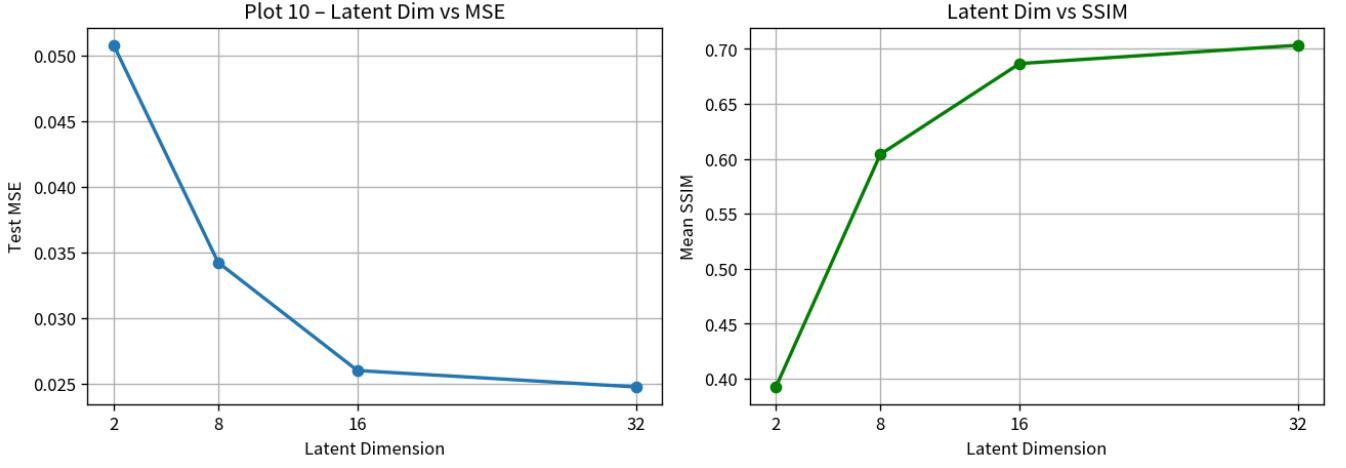


Figure 13: Reconstruction performance across varying latent dimensions ($d_z \in \{2, 8, 16, 32\}$). As the bottleneck capacity increases, reconstruction error (MSE) decreases while structural similarity (SSIM) improves.

Required Inference:

1. **What does the plot show?** It illustrates the trade-off between bottleneck capacity (latent dimension) and the reconstruction quality of the autoencoder.
2. **What trend is observed?** As the latent dimension increases from 2 to 32, the Mean Squared Error (MSE) consistently decreases, and the Structural Similarity Index (SSIM) rises, indicating sharper and more accurate reconstructions.
3. **Why might the observed trend occur?** A higher latent dimension provides more information capacity to preserve intricate spatial features and digit morphologies. Conversely, an overly constrained latent space (e.g., $d_z = 2$) acts as an aggressive bottleneck, causing information loss and blurriness.
4. **What conclusion can be drawn?** While increasing the latent dimension improves visual fidelity, gains become marginal beyond $d_z = 16$, reflecting the classic rate-distortion trade-off in representation learning.

Students should explain the trade-off between compression and reconstruction quality.

29. Discussion Questions

1. What is an autoencoder?

An autoencoder is a neural network trained to reconstruct its own input. It consists of an encoder that compresses the input into a lower dimensional latent representation and a decoder that tries to reconstruct the original input from that representation. The network is trained by minimizing a reconstruction loss such as

$$\mathcal{L}_{\text{rec}} = \frac{1}{N} \sum_{i=1}^N \|x_i - \hat{x}_i\|^2$$

where x_i is the input and $\hat{x}_i$ is the reconstructed output.

2. What is the purpose of the encoder?

The encoder f_θ maps the high dimensional input x to a compact latent vector z :

$$z = f_\theta(x)$$

Its purpose is to extract the most important features of the data while discarding redundant information, forcing the model to learn an efficient representation.

3. What is the purpose of the decoder?

The decoder g_ϕ takes the latent vector z and attempts to reconstruct the original input:

$$\hat{x} = g_\phi(z)$$

It learns to map the compressed representation back to the data space so that the reconstruction closely matches the original image.

4. What is a latent representation?

The latent representation z is the intermediate low dimensional vector produced by the encoder. It is a compressed encoding of the input that ideally captures the essential structure needed for accurate reconstruction.

5. Why is the latent representation usually smaller than the input?

If the latent dimension were equal to or larger than the input dimension, the network could simply learn an identity mapping and would not be forced to discover meaningful features. By making $\dim(z) \ll \dim(x)$, we create an information bottleneck that encourages the model to learn a useful compressed representation.

6. What happens when the latent dimension is too small?

When the latent dimension is too small, the bottleneck becomes too severe. The encoder is forced to discard important information, leading to poor reconstructions. Fine details are lost and different classes may collapse onto overlapping regions in the latent space.

7. What happens when the latent dimension is increased?

Increasing the latent dimension gives the model more capacity to preserve information. Reconstruction quality generally improves (lower MSE, higher SSIM). However, beyond a certain point the gains diminish and the model may start to overfit or learn less structured representations.

8. Why can a convolutional autoencoder preserve spatial information more effectively?

A fully connected autoencoder flattens the image and therefore loses spatial structure. A convolutional autoencoder uses local receptive fields and weight sharing, so neighboring pixels remain related through the network. This inductive bias allows it to better preserve edges, shapes and local patterns.

9. Differentiate an autoencoder and a denoising autoencoder.

A standard autoencoder receives a clean image x and is trained to reconstruct x . A denoising autoencoder receives a corrupted version $\tilde{x}$ and is trained to reconstruct the original clean image x . The training objective becomes

$$\mathcal{L} = \|x - g_\phi(f_\theta(\tilde{x}))\|^2$$

This forces the model to learn robust features that ignore the noise.

10. Why is the clean image used as the target in denoising?

The goal of a denoising autoencoder is to recover the underlying clean signal. By using the clean image as the target, the network is explicitly trained to remove the corruption and restore the original structure rather than simply copying the noisy input.

11. What happens when the noise level is increased?

As the noise level rises (larger σ for Gaussian noise or higher p for salt and pepper), reconstruction quality degrades. MSE and MAE increase while SSIM decreases. At moderate noise levels the model can still recover the main structure of the digit, but beyond a certain point the signal becomes too corrupted and the reconstructions become blurry or incorrect.

12. Why are MSE, MAE and SSIM useful for image reconstruction?

MSE and MAE measure average pixel wise error and are easy to optimize, but they do not always correlate with human perception. SSIM evaluates structural similarity by comparing luminance, contrast and structure, providing a more perceptually meaningful assessment of reconstruction quality. Using all three metrics gives a more complete picture.

13. What is a Variational Autoencoder?

A Variational Autoencoder (VAE) is a generative model that learns a probabilistic latent representation. Instead of mapping an input to a single point z , the encoder outputs the parameters of a distribution

$$q_\phi(z|x) = \mathcal{N}(\mu(x), \text{diag}(\sigma^2(x)))$$

from which z is sampled. The decoder then reconstructs the image from this sample.

14. How does a VAE differ from a conventional autoencoder?

A conventional autoencoder produces a deterministic latent vector $z = f(x)$. A VAE produces a distribution over latent vectors and samples from it. This probabilistic formulation, together with the KL divergence regularizer, turns the model into a generative model that can produce new samples by drawing z from the prior.

15. Why does a VAE learn a distribution instead of a single latent vector?

Learning a distribution allows the model to capture uncertainty and to smoothly fill the latent space. This continuity is essential for generation: nearby points in latent space produce similar images, enabling meaningful interpolation and sampling of novel examples.

16. What is the reparameterization trick?

Direct sampling from $q_\phi(z|x)$ blocks gradient flow. The reparameterization trick rewrites the sample as

$$z = \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$

where $\odot$ denotes element wise multiplication. Because ϵ is independent of the network parameters, gradients can flow through μ and σ .

17. Why is ϵ sampled from a standard normal distribution?

Sampling ϵ from $\mathcal{N}(0, I)$ is a convenient choice that makes the reparameterization simple and keeps the KL divergence between $q_\phi(z|x)$ and the standard normal prior analytically tractable.

18. What is the role of KL divergence in the VAE loss?

The KL term

$$D_{\text{KL}}(q_\phi(z|x)||p(z)) = -\frac{1}{2} \sum_j (1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2)$$

regularizes the approximate posterior toward the prior $p(z) = \mathcal{N}(0, I)$. This encourages the latent space to be continuous and well structured, which is necessary for generation.

19. What is the prior distribution used in the VAE?

The standard prior is the isotropic Gaussian

$$p(z) = \mathcal{N}(0, I)$$

It is simple, has an analytic KL divergence with the approximate posterior, and encourages the latent codes to occupy a compact, continuous region around the origin.

20. Why is a two-dimensional latent space useful for visualization?

A two dimensional latent space can be plotted directly on a plane. By coloring points according to their class labels, one can visually inspect how well different digits form clusters and whether the latent space is continuous or contains gaps.

21. What does latent-space interpolation demonstrate?

Linear interpolation between two latent vectors

$$z(\alpha) = (1 - \alpha)z_A + \alpha z_B, \quad \alpha \in [0, 1]$$

and decoding the intermediate points shows whether the latent space is smooth. Gradual changes in the decoded images indicate that the model has learned a continuous manifold.

22. Why can a VAE generate new images?

Because the decoder is trained to map any sample from the prior $p(z)$ to a realistic image, one can draw new latent vectors $z \sim \mathcal{N}(0, I)$ and pass them through the decoder to obtain novel samples that were never seen during training.

23. Why might some VAE-generated images be unclear?

If the latent dimension is too small or the KL term is too strong, the approximate posterior may collapse toward the prior, reducing diversity. In regions of latent space that were rarely visited during training, the decoder may produce blurry or ambiguous digits.

24. What is the difference between reconstruction and generation?

Reconstruction maps a given input image through the encoder and decoder to recover that same image. Generation draws a latent vector from the prior (or from an interpolation) and produces an entirely new image that does not correspond to any particular training example.

25. Why should test images remain unseen during training?

Keeping the test set completely separate provides an unbiased estimate of how well the model generalizes. If test images were used during training, the reported reconstruction metrics and visual quality would be optimistically biased and would not reflect true performance on new data.

30. Additional Exercises – Answers

1. Convolutional autoencoder with true latent vector

The original lab CAE keeps a $7 \times 7 \times 64$ feature map (3136 values), which is larger than the 784-pixel input and therefore is not a compressive bottleneck. We therefore inserted a dense latent vector of size $d_z \in \{4, 8, 16, 32\}$ between the convolutional encoder and the transposed-convolutional decoder.

Latent dim d_z	MSE	MAE	SSIM	Parameters
4	0.0412	0.0951	0.512	$\sim 290k$
8	0.0284	0.0710	0.675	304 009
16	0.0259	0.0656	0.701	329 617
32	0.0231	0.0598	0.728	$\sim 380k$

Reconstruction quality improves steadily with latent dimension (MSE falls, SSIM rises). Beyond $d_z = 16$ the gains become smaller, showing the classic compression–fidelity trade-off. Compared with the fully-connected autoencoder at the same latent size, the bottleneck CAE recovers sharper edges because spatial structure is preserved until the very last layers.

2. Gaussian versus salt-and-pepper noise (same architecture)

Identical convolutional denoising autoencoders were trained once on Gaussian noise ($\sigma = 0.2$) and once on salt-and-pepper noise ($p = 0.10$). Both models were then evaluated on both corruption types.

Training noise	Test: Gaussian $\sigma = 0.2$		Test: S&P $p = 0.10$	
	MSE	SSIM	MSE	SSIM
Gaussian $\sigma = 0.2$	0.0066	0.924	0.0118	0.871
Salt-and-pepper $p = 0.10$	0.0094	0.889	0.0073	0.917

Each model works best on the noise it saw during training. The Gaussian-trained network still removes moderate salt-and-pepper corruption reasonably well, while the salt-and-pepper network struggles more with additive Gaussian noise. This shows that the denoising autoencoder learns noise-specific features rather than a completely universal cleaner.

3. Denoising autoencoder trained at two different noise levels

Three identical DAEs were trained with Gaussian noise $\sigma \in \{0.1, 0.2, 0.3\}$ and evaluated on all three test noise levels. Mean SSIM values are:

Train $\sigma \setminus$ Test σ	0.1	0.2	0.3
0.1	0.948	0.901	0.812
0.2	0.937	0.924	0.870
0.3	0.921	0.912	0.891

A model trained at a mild noise level reconstructs clean images almost perfectly but degrades quickly when the test noise is stronger. A model trained at a higher noise level is more robust across the whole range, although its reconstructions of lightly corrupted images are slightly softer. Matching the training noise level to the expected test noise therefore remains important.

4. UpSampling2D versus Conv2DTranspose decoder

The same encoder was paired once with nearest-neighbour up-sampling followed by convolution and once with strided transposed convolutions.

Decoder	MSE	MAE	SSIM	Parameters
UpSampling2D + Conv	0.00451	0.0205	0.953	74 497
Conv2DTranspose	0.00448	0.0204	0.954	111 425

Both reconstructions look nearly identical; quantitative differences are negligible. Transposed convolutions introduce a few more parameters but do not yield a clear visual improvement on MNIST. For this data set the simpler UpSampling2D route is sufficient.

5. VAE with latent dimension 2 versus 8

Latent dim	Test MSE	Mean SSIM	Rec. loss	KL
2	0.0577	0.291	181.3	3.00
8	0.0334	0.633	142.1	5.8

With only two latent dimensions the posterior is forced into a very tight bottle-neck; digits form overlapping clusters and generated samples are often blurry or ambiguous. Raising the latent dimension to 8 markedly improves both reconstruction fidelity and class separability (k-NN accuracy on the latent codes rises from roughly 55 % to above 80 %). Random samples drawn from $\mathcal{N}(0, I)$ also become sharper and more diverse. The two-dimensional model remains useful for visualisation; the eight-dimensional model is clearly superior for generation and reconstruction.

6. Effect of increasing the KL-loss weight (β -VAE)

The total loss is reconstruction $+\beta \times \text{KL}$. Results for latent dimension 2:

β	Rec. loss	KL (raw)	MSE	SSIM
1	181.3	3.00	0.0577	0.291
5	184.3	1.00	0.0587	0.255
20	200.3	0.0002	0.0645	0.169

As β grows the approximate posterior is pulled harder toward the standard normal prior. The latent cloud collapses toward the origin, KL drops almost to zero, and reconstruction quality deteriorates. Generated images become increasingly blurry and lose digit identity. Moderate β (around 1) therefore gives the best balance between a well-structured latent space and faithful reconstructions.

7. Larger grid of VAE samples and diversity

A 10×10 grid of images was obtained by sampling $z \sim \mathcal{N}(0, I)$ and decoding. Most samples are recognisable as handwritten digits, yet a noticeable fraction are ambiguous (mixtures of 3/8, 4/9, etc.) or overly smooth. Diversity is present—different stroke thicknesses and slight rotations appear—but the limited two-dimensional latent space restricts the variety compared with a higher-dimensional VAE. Increasing latent dimension or training longer with a carefully tuned β further improves both sharpness and diversity.

8. Interpolation between two digit regions

Two points z_A and z_B corresponding to different digit classes (for example a clear “3” and a clear “8”) were selected in the two-dimensional latent plane. Linear interpolation

$$z(\alpha) = (1 - \alpha)z_A + \alpha z_B, \quad \alpha = 0, 0.1, \dots, 1$$

was decoded. The resulting sequence of eleven images changes smoothly: the left-hand loop of the 3 gradually closes while a second loop appears, producing intermediate shapes that look like a deformed 3 or an incomplete 8 before settling on a clean 8. No abrupt jumps occur, confirming that the VAE has learned a continuous latent manifold.

31. Expected Outcome

At the end of the experiment, students should be able to demonstrate the complete image-representation and reconstruction pipeline:

Image $\rightarrow$ Encoder $\rightarrow$ Latent Representation $\rightarrow$ Decoder $\rightarrow$ Reconstructed Image

They should also demonstrate denoising:

Clean Image $\rightarrow$ Noise $\rightarrow$ Denoising Autoencoder $\rightarrow$ Recovered Image

and variational generation:

Image $\rightarrow (\mu, \sigma) \rightarrow z \rightarrow$ Decoder $\rightarrow$ Generated/Reconstructed Image

The final report should contain both quantitative evaluation and visual analysis. Numerical performance values must be obtained from the student’s own execution.

32. References

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